NG YOU JUIN

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PROFESSIONAL SUMMARY

Nanyang Technological University (NTU) Computer Science undergraduate (CGPA: 4.01/5.00) with practical software engineering experience across full-stack web applications, RESTful microservice architectures, and operational automation. Proven track record at GovTech engineering production web features using React, Node.js, Express, and MongoDB, and architecting high-performance hackathon solutions at DSTA BrainHack leveraging Next.js, Supabase (PostgreSQL), JWT authentication, and Groq LLM pipelines. Strong foundation in data structures, algorithms, and secure backend systems developed through national service at The Digital Intelligence Service (DIS).

TECHNICAL SKILLS & COMPETENCIES

- **Programming Languages:** TypeScript, JavaScript, Python, Java (OOP, Concurrency), C, C++, SQL, Bash
- **Frontend Engineering:** React, Next.js, HTML5, CSS3, Tailwind CSS, Responsive UI Design, Client-Side Caching
- **Backend & APIs:** Node.js, Express.js, RESTful API Design, Microservices, Next.js API Routes, JWT Authentication, API Security Middleware
- **Databases & Cloud:** PostgreSQL, Supabase (RLS), MongoDB, MySQL, SQLite, Firebase, Redis (Basics), Vercel
- **DevOps & Engineering Practices:** Git, GitHub, CI/CD Fundamentals, Linux/Unix Shell, Postman, Agile/Scrum, Software Quality Assurance (SQA)

WORK & INTERNSHIP EXPERIENCE

GovTech Singapore — *Software Engineer Intern — Smart Facilities Management Platform* **Apr 2021 – Jun 2022**

- **Full-Stack Feature Engineering:** Designed and built modular frontend and backend features for an enterprise smart-facilities management web platform using React, Node.js, Express, and MongoDB.
- **Component & Form Integration:** Embedded Form.io management workflows via isolated iframes and secure cross-frame event communication, eliminating cross-domain redirection while preserving transactional session state.
- **Root-Cause Troubleshooting & Defect Resolution:** Investigated software defects, asynchronous race conditions, and bottlenecks through systematic code analysis, delivering verified performance patches across Agile sprints.

The Digital Intelligence Service (DIS) – Singapore Armed Forces — *Cyber Security Specialist & Logistics Administrator* **Sep 2023 – Jul 2025**

- **Workflow Digitisation & Automation:** Conceptualised, developed, and deployed a barcode-enabled asset management application using Microsoft Power Apps and Power Automate, eliminating paper-based logs and accelerating equipment handovers.
- **Vulnerability Assessment & Hardening:** Executed structured vulnerability evaluations across operational environments, identifying configuration and software flaws to deliver actionable hardening recommendations.
- **Operational Reliability & Audit Tracking:** Supervised critical technical equipment inventory, validated database records, and coordinated procurement pipelines to ensure continuous audit compliance.

SOFTWARE ENGINEERING & AI PROJECTS

DSTA BrainHack Hackathon – Finalist | *Backend & AI Integration Developer | Next.js, Supabase, PostgreSQL, Groq API* **May 2026 – Jun 2026**

- **Scalable Backend & Schema Design:** Engineered backend API routes and modeled normalized relational data in Supabase (PostgreSQL) to handle user state, scam submissions, and real-time database transitions.
- **Security & Middleware:** Implemented stateless JWT-based authentication and custom Next.js middleware to validate request headers, verify cryptographic tokens, and protect sensitive routes against unauthorized access.
- **Concurrency & Anti-Abuse Logic:** Designed deterministic community-voting algorithms to prevent duplicate voting, handle concurrent client requests, compute real-time scores, and disburse engagement points upon poll close.
- **LLM Pipeline Integration:** Integrated Groq's high-throughput LLM API to perform automated text extraction, category prediction, and suspicious keyword detection from unstructured text submissions.

EDUCATION

Nanyang Technological University (NTU), Singapore — *Bachelor of Computing in Computer Science (CGPA: 4.01 / 5.00)* **Jul 2025 – Jul 2029 (Expected)**

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Design and Programming (Java), Software Engineering, Operating Systems (Process Scheduling, Concurrency), Computer Networks.

Temasek Polytechnic (TP), Singapore — *Diploma in Information Technology (CGPA: 3.78 / 4.00)* **Feb 2019 – Apr 2022**

- **Relevant Coursework:** Full Stack Web Development, Microservices Architecture, Software Quality Assurance, Data Structures and Algorithms, Object-Oriented Analysis and Design, Mobile App Development.